

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The data warehouse tools are...



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.
Understand the concept of data warehouse, its components and characteristics.

Can Do Statements:

- I can talk about data warehouse tools and utilities.
- I can give information about data warehouse architecture properties.

Ask your partner:

- Do you know the different tools data warehouse possess? Why are they important?
- What are the properties of data warehouse architecture?





Let's Explore!

These are the tools data warehouse has.



Instructions: Read the text below.

Today we are going to cover two important topics related to data warehouse: Tools and utilities and architecture. We are not going to go over other aspects of data warehouse during this session as I expect to provide detailed information on the two topics I just mentioned.

I'm starting with tools and utilities. We are going to use five different tools and utilities when working with data warehouse. The first one is data extraction, which refers to the gathering of data from various sources. We are not going to limit ourselves to using only homogeneous sources as a larger variety of these will make our analysis more complete. The second one is data cleaning that we will use when finding and correcting mistakes in data. Thirdly, we have data transformation which refers to the process of converting data from a legacy format to a warehouse one. Then, we are using data loading when sorting, consolidating, summarizing and building indices and partitions in order to make data more comprehensible and practical to use. Finally, we are going to work with refreshing which refers to the process of updating data sources to warehouses.

Now, for our second topic, data warehouse architecture, we are going to go over five different properties. The first one is separation, which we are going to consider when handling data since analytical and transactional processing must remain apart. The second one, scalability, refers to the easy upgrade of hardware and software architectures. This is due to the fact that our business is not going to be at a standstill, but is expected to have a progressive increase of the data volume and the number of user's requirements. As a third property, we will have extensibility which determines that we are not going to have to restructure the whole system when the architecture includes new technologies and applications. Fourthly, security refers to the importance of the monitoring accesses as there is strategic data stored in data warehouses and knowing who accesses this information will ensure that data remains safe. Finally, we are going to consider administerability as data warehouse management mustn't be too complex or hard to handle.

Alright, I'm passing the list of articles you'll have to read for next session. See you on Monday and the class is dismissed.





Let's Practice!

The students go over data warehouse tools and utensils before next session.

Instructions: Write the correct word to complete each sentence Use the following words: cleaning, extraction, loading, Refreshing, transformation.

1. I'm not going to be able to use the _____ tool properly if I don't gather information from different sources.
2. The data warehouse is not going to show any errors in the data as we've been very careful with the _____ tool.
3. _____ is the tool or utensil we use when updating data sources to warehouses.
4. I'm learning how to transform data from legacy to warehouse format. That's known as _____ tool, right?
5. I am going to need you to teach me the _____ process, I can't remember how to consolidate and summarize indices and partitions.



Let's Practice!

Data Warehouse Characteristics

Instructions: Choose the right answer to complete each statement.

1. Separation is the data warehouse property where...
(HINT: Separation refers to the clear division of two types of processing.)
 - a. analytical and transactional processing must stay apart.
 - b. software and hardware upgrading must be as easy as possible.
 - c. new technologies are integrated without having to modify the whole system.
 - d. all of the processes and data available need to be tracked at all times.
2. I will tell you that scalability states that...
(HINT: Scalability refers to the process of increasing or updating something progressively.)
 - a. data warehouse management must not be too difficult to handle.
 - b. software and hardware upgrading must be as easy as possible.
 - c. new technologies should be integrated to the architecture without having to change the whole system.
 - d. all of the processes and data available need to be tracked at all times.



**Let's Practice!**

3. I'm going to explain to my classmates what the architecture of a data warehouse is, so I read that extensibility is the property where...
 - a. (HINT: Extensibility refers to the feature of being able to hold new changes without having to restructure the existing system.)
 - b. analytical and transactional processes must be apart.
 - c. data warehouse management must not be too difficult to handle.
 - d. new technologies and applications can be integrated without having to change the whole system.
 - e. all of the processes and data available need to be tracked at all times
4. I'm reading that security is the property which states that...
(HINT: It is important to keep track of who accesses data in order to assure that information is kept safe.)
 - a. analytical and transactional processes must stay apart.
 - b. software and hardware upgrading must be as easy as possible.
 - c. new technologies and applications can be integrated without having to change the whole system.
 - d. all of the processes and data available need to be tracked constantly.
5. Administerability is the property which determines that...
 - a. data warehouse management must not be too difficult to handle.
 - b. software and hardware upgrading must be as easy as possible.
 - c. new technologies can be easily integrated.





Extension Activity

How does a company uses the data warehouse architecture and tools?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Think of a company that uses Data Warehouse Architecture.
- Look for the tools and architecture properties that company is using.
- Compare them with the tools and properties you know.
- Write a chart comparing the tools and properties described in this lesson with the ones the company is using.
- Share your work with your partners.



Target Words

- homogeneous** (adjective): referring to a group of elements that have the same characterizes and do not differ from each other
e.g. We are not going to limit ourselves to using only homogeneous sources as a larger variety of these will make our analysis more complete.
- legacy format** (noun): it is an old format for storing data that is difficult to comprehend and use
e.g. Thirdly, we have data transformation which refers to the process of converting data from a legacy format to a warehouse one.
- standstill** (noun): the cessation of progress or activity
e.g. The second one, scalability, refers to the easy upgrade of hardware and software architectures. This is due to the fact that our business is not going to be at a standstill, but is expected to have a progressive increase of the data volume and the number of user's requirements.

Self-Assessment

Yes

No

- I find a company that uses Data Warehouse Architecture
- I look for the tools and architecture properties that company is using
- I compare them with the tools and properties of the lesson.
- I write a chart comparing both.
- I share my work with partners.

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- partition** (noun): an element or group of elements belonging to a wider system that are differentiated by having certain characteristics
e.g. We are using data loading when sorting, consolidating, summarizing and building indices and partitions in order to make data more comprehensible and practical to use.
- the class is dismissed** (expression): the indication that students can leave the classroom when the lesson has finished
e.g. I'm passing the list of articles you'll have to read for next session. See you on Monday and the class is dismissed.
- sort** (verb): to separate or classify according to certain established parameters
e.g. We are using data loading when sorting, consolidating, summarizing and building indices and partitions in order to make data more comprehensible and practical to use.



Learning Points

The Future: Going To – Negative Statements

You use **going to** to talk about future plans or intentions that have been planned or prepared before.

To create negative statements using **going to** you need to use the following grammar structure:

Subject + **verb to be** + **not** + **going to** + **main verb in its base form** + complement

Example:

*She **is not going to fail** the test.*

*I'm **not going to work** tonight at the warehouse.*

*They **are not going to store** all their products in their warehouses.*

Examples

- *We are not going to go over other aspects of data warehouse during this session as I expect to provide detailed information on the two topics I just mentioned.*
- *The first one is data extraction, which refers to the gathering of data from various sources. We are not going to limit ourselves to using only homogeneous sources as a larger variety of these will make our analysis more complete.*

Data Warehouse Architecture

There are some important properties that build **data warehouse architecture**.

We can define **data warehouse** as the gathering of information that helps managers, directors and analysts to make certain decision in the management of a warehouse. Let's take a look at the five essential properties of data warehouse architecture.

Separation: It is the property which determines that analytical and transactional processing must remain apart.

Scalability: It is the property which states that hardware and software architectures must be easy to upgrade or improve. This is due to the fact that the data volume and the number of user's requirements are increasing progressively.

Extensibility: It is the property which determines that the architecture of a warehouse must be able to include or work with the technologies and applications without having to restructure the system.

Security: This property states that monitoring access is vital as there is strategic data stored in data warehouses.

Administerability: It is the property which defines that data warehouse management shouldn't be too complicated.

